

A2A Knowledge Mesh

Catch enterprise knowledge drift before it catches you.

Created by: Eliott Raguin & Maël Perrigaud

Scraper

Data Extraction

Keeper

Data Storage

Reconciler

Conflict Resolution

Registry

Data Indexing

Bridge

System Overview

Five specialized agents · One Band room · Real-time drift detection

THE CHALLENGE

Enterprise Knowledge Drift

Docs say one thing. Code says another. CI lives in its own world. Nobody notices until it breaks.

README

Python 3.9

pip install

pyproject.toml

Python >=3.11

uv install

CI config

Python 3.12

Dockerfile

Python 3.10

Architecture docs

Firebase auth

Code base

Supabase auth

⚠ Same project. Six contradicting sources. Which one is the actual ground truth?

THE SOLUTION

Five Agents, One Room

Band-native multi-agent platform that detects, scores, and resolves knowledge drift in real time.

Scraper

Reads repos, extracts structured facts via LLM

Keeper

SQLite fact store with $O(n \log n)$ conflict detection via JOIN

Reconciler

LLM-powered conflict scoring + interactive resolution loop

Registry

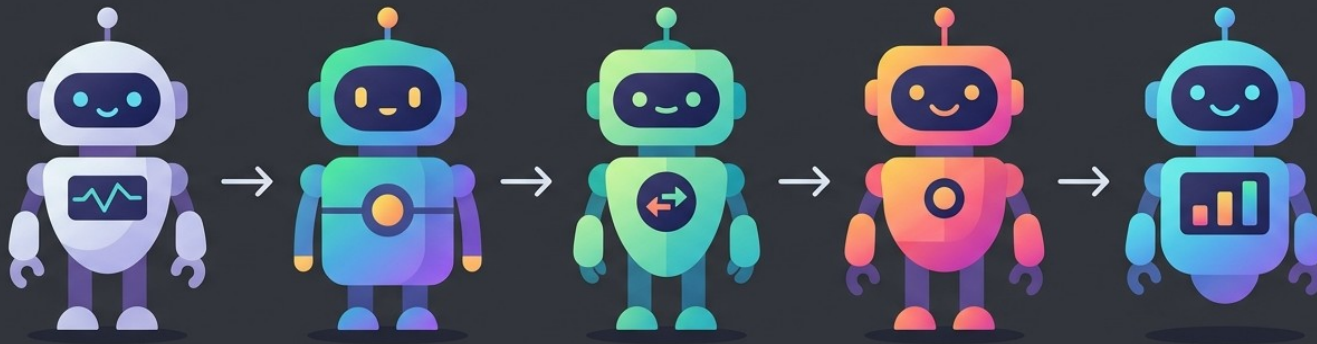
Dynamic service discovery for mesh agents

Bridge

Real-time dashboard with modern glassmorphism UI

Collaborative Multi-Agent Flow

A2A KNOWLEDGE MESH



Scraper:
Extracts facts

Keeper:
Stores facts

Reconciler:
Resolves conflicts

Registry:
Agent directory

Bridge:
Live dashboard

Key Infrastructure Details

- Band Room Transport: agents communicate using @mentions and WebSockets.
- Decentralized DB: each agent maintains its own SQLite database.
- A2A Protocol: standard JSON-RPC 2.0 endpoints for local agent requests.
- Provenance Tracked: facts are RDF-lite triples containing subject, predicate, object, source_id, and timestamp.

Secure bearer authentication + HMAC payload signatures protect all inter-agent operations.

DEMONSTRATION

Watch It In Action

1

Scraper scans repo

8 facts extracted from docs-repo + code-repo

2

Keeper detects conflicts

3 contradictions found (framework, version, database)

3

Reconciler scores severity

critical · high · medium — with LLM confidence ratings

4

Human reviews & resolves

Interactive loop via @mentions in Band room

Dashboard updates live

Timeline, counters, and mesh graph reflect changes

Run: `bash scripts/run_mesh.sh` · Offline mode: `uv run python scripts/quick_demo.py`

Knowledge Mesh Graph

Subject node

Valid fact

Active conflict

- mesh graphify generates an interactive Vis.js network graph
- Click any node to inspect: versions, timestamps, source files
- Subjects (blue) connect to facts (green) — conflicts (red) are visually scannable
- Zoom, pan, filter — built for human-in-the-loop debugging



Technology Architecture

Runtime & Storage

Python 3.11+ · Starlette + Uvicorn · SQLite WAL mode

Agent Coordination

Band SDK (WebSocket) · A2A Protocol (JSON-RPC 2.0)

LLM Inference

Featherless / OpenAI API · Resilience: 2x retries + fallback

Security & Auth

HMAC body signing · Per-role bearer tokens · master fallback

Developer Tools

Glassmorphic dashboard UI · Interactive Vis.js network graph

Testing & Fixtures

std unittest · Offline fixtures covering 4 drift scenarios

Why Band of Agents?

- Shared room collaboration: 5 agents communicate seamlessly in one room.
- Structured context hand-off: @mentions transfer structured fact payloads.
- Real-time events: WebSockets keep system dashboard synchronized without polling.
- Human-in-the-loop: Reconciler offers AI recommendations, human stays in control.
- Open standard ready: standard LVK/WebSocket and local A2A fallback options.
- Single-command launch: one unified script starts all agents and database services.

Market Scope & Value Proposition

Market Size & Opportunity

- TAM: \$12B+ Enterprise Knowledge Management Market.
- SAM: \$1.2B DevOps & CI/CD Integrity Assurance sector.
- Revenue Stream 1: SaaS subscriptions per repository & active user.
- Revenue Stream 2: Self-hosted enterprise licenses for high security compliance.
- High Growth: Drift detection is critical for companies utilizing AI and automated pipelines.

Unique Selling Proposition (USP)

- Active Reconciliation: Unlike passive sync tools (e.g. Memory Store, YC P26) which copy raw chat logs, we check all files and detect actual code vs doc contradictions.
- Conflict Scoring: Automatically scores drift severity using local LLM algorithms.
- Human-in-the-loop: Allows easy resolution through Band room commands, keeping developers in control.
- Full Provenance: Tracks source_id, timestamp, and version for every fact and conflict resolved.

A2A Knowledge Mesh

github.com/maelemiel/a2a-knowledge-mesh

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Built for Band of Agents Hackathon 2026